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Synthetic Signature Generation for Online Biometric Verification

Evaluation of Synthetic Data for Signature Enrollment Using DTW and Statistical Classifiers

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Finally, I would like to thank Polytech Clermont for giving me the opportunity to carry out this international research internship. This experience allowed me to deepen my knowledge in machine learning, biometric systems, and scientific research while developing both my technical and analytical skills.

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1. ABSTRACT

Online handwritten signature verification is a widely used biometric authentication technique. Most verification systems require several genuine signatures from each user during the enrollment phase in order to accurately model intra-class variability. However, collecting a sufficient number of genuine signatures is often difficult in real-world applications.

The objective of this internship was to investigate whether synthetic signatures generated by deep learning could be used to augment the enrollment dataset while maintaining the performance of signature verification systems. To address this problem, an autoencoder-based generative model was developed to produce synthetic online signatures from genuine samples contained in the MCYT signature database.

The generated signatures were first evaluated using Dynamic Time Warping (DTW), a reference method for comparing temporal sequences. Following the recommendations of the research supervisor, the evaluation was then extended to statistical classifiers. A Random Forest classifier was initially implemented before reproducing a diagonal Linear Discriminant Analysis (diagLDA) classifier based on a 230-dimensional handcrafted feature representation.

Several enrollment scenarios were compared, including one genuine signature, five genuine signatures, and one genuine signature augmented with synthetic signatures. Performance was assessed using the Identification Rate, Equal Error Rate (EER), and minimum Detection Cost Function (minDCF).

The experimental results showed that the baseline using five genuine enrollment signatures remained the most effective configuration. Although the generated signatures exhibited realistic characteristics, they were not able to fully reproduce the natural intra-user variability required by the statistical classifiers. Consequently, augmenting the enrollment set with synthetic signatures did not improve the verification performance under the evaluated conditions.

This work provides a complete framework for synthetic signature generation and evaluation while highlighting both the potential and the current limitations of deep learning-based data augmentation for online signature verification.

Keywords: Online Signature Verification, Biometrics, Data Augmentation, Deep Learning, Autoencoder, Dynamic Time Warping, Random Forest, diagLDA.

2. INTRODUCTION

2.1 CONTEXT

Biometric authentication has become an essential component of modern security systems. Unlike traditional authentication methods based on passwords or identification cards, biometric systems rely on unique physiological or behavioral characteristics to verify an individual's identity. Among the different biometric modalities, online handwritten signatures remain particularly attractive because they are non-invasive, widely accepted by users, and legally recognized in many administrative and financial applications.

Online handwritten signatures differ from static handwritten signatures because they capture the complete writing process rather than only its final appearance. During acquisition, a digitizing device records dynamic information such as the pen trajectory, writing speed, pressure, and timing. These temporal characteristics provide valuable information for distinguishing genuine signatures from forged ones and significantly improve verification performance.

As a result, online signature verification has become an active research field at the intersection of biometrics, machine learning, and pattern recognition. Numerous verification approaches have been proposed over the years, ranging from sequence comparison algorithms to modern statistical and deep learning techniques.

2.2 PROBLEM STATEMENT

Despite the significant progress achieved in online signature verification, one important limitation remains during the enrollment phase of a biometric system.

Most verification methods require several genuine signatures from each user to accurately model the natural variability of their handwriting. However, collecting a large number of signatures is often impractical in real-world applications, as users generally provide only a limited number of enrollment samples.

This limitation raises an important research question:

Can synthetic signatures generated by artificial intelligence compensate for the lack of genuine enrollment signatures while maintaining the performance of signature verification systems?

Answering this question requires not only generating visually realistic signatures but also ensuring that the generated samples preserve the discriminative characteristics needed by verification algorithms.

2.3 OBJECTIVES OF THE INTERNSHIP

The main objective of this internship was to investigate the contribution of synthetic signature generation to online handwritten signature verification.

More specifically, the work focused on the following objectives:

- understanding the structure and characteristics of the MCYT online signature database;
- developing a deep learning model capable of generating realistic synthetic signatures from genuine samples;
- evaluating the quality of the generated signatures using Dynamic Time Warping (DTW), a well-established sequence comparison algorithm;
- studying whether synthetic signatures can improve the enrollment process when used with statistical classifiers;
- comparing different enrollment configurations involving genuine and synthetic signatures through several performance metrics.

Throughout the internship, the project progressively evolved from a signature generation problem toward a complete evaluation framework combining deep learning, traditional sequence matching techniques, and statistical classification methods.

3. HOST INSTITUTION AND RESEARCH ENVIRONMENT

3.1 HOST INSTITUTION

This internship was carried out at TecnoCampus Mataró-Maresme, a university and technology park located in Mataró, Spain, approximately 30 kilometers north of Barcelona. TecnoCampus is a higher education institution affiliated with Pompeu Fabra University (UPF) and combines academic education, applied research, innovation, and entrepreneurship within the same environment, promoting close collaboration between students, researchers, and companies.

The institution hosts several research groups working in fields such as engineering, artificial intelligence, healthcare technologies, and digital innovation. This multidisciplinary environment provides students with the opportunity to participate in applied research projects while developing practical and scientific skills.

3.2 RESEARCH GROUP

The internship was conducted within the Signal and Data Processing Research Group (TSD) under the supervision of Professor Marcos Faúndez-Zanuy.

The group's research focuses on signal processing, machine learning, and biometric systems, with particular expertise in online handwritten signature verification **[1].** Its work includes the development of authentication methods based on behavioral biometrics as well as the application of artificial intelligence techniques to biometric recognition.

Working within this research group provided an excellent opportunity to discover how machine learning techniques are applied to real biometric verification problems.

3.3 INTERNSHIP CONTEXT

The objective of the internship was to study whether synthetic online handwritten signatures generated by deep learning could improve the enrollment stage of biometric signature verification systems.

The project began with the analysis of the MCYT online signature database and the understanding of existing verification methods, particularly Dynamic Time Warping (DTW). An autoencoder was then developed to generate synthetic signatures from genuine samples.

To evaluate their usefulness, different verification approaches were investigated. The generated signatures were first compared using DTW before being evaluated with statistical classifiers, including Random Forest and diagonal Linear Discriminant Analysis (diagLDA).

The experiments compared several enrollment scenarios combining genuine and synthetic signatures in order to assess the impact of data augmentation on verification performance.

4. STATE OF THE ART

4.1 BIOMETRICS

Biometric authentication has become an increasingly important research field due to the growing demand for secure and reliable identity verification systems. Unlike traditional authentication methods based on passwords or physical tokens, biometric systems rely on unique physiological or behavioral characteristics that are difficult to reproduce or steal.

Biometric traits can generally be divided into two categories. Physiological biometrics include physical characteristics such as fingerprints, iris patterns, facial features, and palmprints. These traits remain relatively stable throughout an individual's lifetime. In contrast, behavioral biometrics analyze the way a person performs a specific action, such as speaking, typing on a keyboard, walking, or signing their name.

Among behavioral biometrics, handwritten signatures occupy a particular position. Unlike many other biometric traits, signatures have long been accepted as a legal proof of identity in administrative, financial, and commercial contexts. Their familiarity and social acceptance make them an attractive authentication method despite the challenges associated with their variability.

For these reasons, handwritten signature verification continues to attract significant research interest, particularly with the emergence of digital devices capable of recording dynamic writing information **[2].**

4.2 ONLINE HANDWRITTEN SIGNATURE VERIFICATION

Signature verification systems can be divided into two categories: offline and online verification.

Offline verification is based solely on the final image of a handwritten signature. Since only static visual information is available, the verification process mainly relies on image processing techniques.

Online signature verification, on the other hand, captures the complete writing process using a digitizing tablet or a touchscreen device. During acquisition, several dynamic signals are recorded, including: horizontal pen position (x-coordinate), vertical pen position (y-coordinate), pen pressure, timestamp and pen-up and pen-down events.

These temporal signals contain valuable behavioral information that cannot be observed in a scanned image **[2].** Consequently, online verification generally provides higher accuracy than offline verification and is more resistant to skilled forgeries.

However, online signatures also present an important challenge: the same individual never produces two perfectly identical signatures. Natural variations in speed, pressure, or writing rhythm introduce intra-user variability, making signature verification a complex pattern recognition problem.

4.3 THE ENROLLMENT PROBLEM

The performance of a biometric verification system strongly depends on the quality of the enrollment phase.

During enrollment, several genuine signatures are collected to create the reference model of each user. These samples are intended to capture the natural variability of the person's handwriting.

In practice, collecting numerous enrollment signatures is often inconvenient. Users are generally willing to provide only a small number of samples, while verification algorithms typically perform better when more genuine signatures are available.

This limitation motivates the development of data augmentation techniques, whose objective is to artificially increase the size of the enrollment dataset while preserving the biometric characteristics of the original user **[3][4].**

Recent advances in deep learning have made it possible to generate synthetic biometric samples that resemble genuine observations. Nevertheless, an important question remains: are these synthetic signatures sufficiently realistic to improve the performance of signature verification systems?

This question constitutes the main motivation of the present internship.

4.4 THE MCYT SIGNATURE DATABASE

The experiments conducted during this internship were based on the MCYT online signature database **[5],** one of the most widely used benchmark datasets in online signature verification research.

The database contains signatures acquired using a digitizing tablet capable of recording both spatial and temporal information. Each signature is represented as a sequence of points describing the pen trajectory over time.

Besides genuine signatures, the database also includes skilled forgeries produced after visual imitation of genuine samples. This characteristic makes MCYT particularly suitable for evaluating biometric verification systems under realistic conditions.

The widespread use of the MCYT database in the scientific literature also facilitates comparisons with previously published methods.

4.5 DYNAMIC TIME WARPING (DTW)

One of the most established methods for comparing temporal sequences is Dynamic Time Warping (DTW) **[6].**

Originally introduced for speech recognition, DTW measures the similarity between two sequences that may differ in duration or execution speed. Instead of comparing points one by one, the algorithm computes an optimal alignment by allowing local stretching and compression of the temporal axis.

This property makes DTW particularly well suited to online handwritten signatures, since two genuine signatures written by the same person rarely have exactly the same length or writing speed.

Given two signatures, DTW computes a cumulative distance representing their overall similarity. Smaller distances indicate more similar signatures, while larger distances suggest greater dissimilarity.

Because of its robustness to temporal distortions, DTW has become one of the reference algorithms in online signature verification and has been extensively used as a baseline in biometric research.

In this internship, DTW was first employed to evaluate the similarity between genuine and synthetic signatures before extending the evaluation to statistical classifiers.

4.6 DEEP LEARNING FOR SYNTHETIC SIGNATURE GENERATION

Deep learning has considerably expanded the possibilities of data generation and augmentation.

Among the different architectures, autoencoders are widely used for unsupervised representation learning. An autoencoder consists of two neural networks:

- an encoder, which compresses the input data into a lower-dimensional latent representation;
- a decoder, which reconstructs the original data from this latent space.

Once trained, the latent representation captures the essential characteristics of the input signatures. By introducing controlled perturbations into the latent space, it becomes possible to generate new signatures that preserve the general writing style of the original user while introducing slight variations.

This principle forms the basis of the synthetic signature generation approach developed during this internship.

4.7 STATISTICAL CLASSIFICATION METHODS

Although DTW provides a direct measure of similarity between two signatures, modern biometric systems often rely on statistical classifiers capable of exploiting high-dimensional feature representations.

During this internship, two statistical approaches were investigated.

The first one was a Random Forest classifier, an ensemble learning method based on multiple decision trees **[7].** Random Forest classifiers are known for their robustness, ease of implementation, and ability to model complex decision boundaries.

The second approach relied on diagonal Linear Discriminant Analysis (diagLDA). This classifier has been extensively used in the literature for online signature verification because of its efficiency when handling handcrafted biometric features **[8].** In this work, each signature was represented by a vector of 230 features describing various dynamic characteristics such as position, velocity, acceleration, pressure, and frequency-domain information **[9].**

These classifiers allowed the evaluation of synthetic signatures under realistic verification conditions rather than relying solely on pairwise similarity measures.

4.8 PERFORMANCE EVALUATION

The effectiveness of a biometric verification system is commonly assessed using several performance metrics.

The Identification Rate (IR) measures the percentage of correctly identified users. A higher Identification Rate indicates better recognition performance.

The Equal Error Rate (EER) corresponds to the operating point where the False Acceptance Rate and the False Rejection Rate are equal. Since it simultaneously reflects both types of classification errors, it is widely used to compare biometric systems.

Another commonly reported metric is the minimum Detection Cost Function (minDCF), which combines different types of verification errors according to predefined application costs. Lower values indicate better overall system performance.

These evaluation metrics were used throughout the internship to compare different enrollment strategies involving genuine and synthetic signatures.

5. METHODOLOGY

This chapter presents the methodology followed during the internship. The work was developed progressively, starting with the preparation and analysis of the MCYT signature database, followed by the development of an autoencoder for synthetic signature generation. The generated samples were first evaluated using Dynamic Time Warping (DTW). In a second stage, the study focused on the use of synthetic signatures as additional enrollment samples. Statistical classifiers were therefore introduced to evaluate whether synthetic data could compensate for a limited number of genuine signatures.

5.1 MCYT DATASET PREPARATION

The experiments were conducted using the MCYT online signature database **[5].** Each signature is stored as a temporal sequence containing information recorded during the signing process. The available signals provide both spatial and dynamic information about the handwriting movement.

Two main types of samples were considered:

- genuine signatures, produced by the legitimate user;
- skilled forgeries, produced by another person attempting to imitate the genuine signature.

Random forgeries were also considered during the evaluation. In this case, a genuine signature from another user is presented as if it belonged to the claimed user.

For the first DTW experiments, five genuine signatures were used to construct the reference model of each user: v01, v02, v03, v04 and v05.

The following five genuine signatures were reserved for testing: v06, v07, v08, v09 and v10.

This separation ensured that the genuine signatures used for evaluation had never been included in the reference model.

The skilled forgery samples were kept exclusively for testing. This allowed the system to be evaluated against signatures specifically designed to imitate the legitimate user's handwriting.

The same general separation between enrollment and evaluation data was maintained throughout the experiments in order to avoid data leakage.

5.2 SIGNATURE PREPROCESSING

Before being used by the neural network, the raw signature sequences had to be transformed into a common representation.

Online signatures naturally have different durations and therefore contain different numbers of points. However, neural networks require inputs with consistent dimensions. Each signature was consequently resampled to a fixed sequence length.

The input representation used for the autoencoder was based on three main dynamic signals:

- horizontal position (x);
- vertical position (y);
- pressure (p).

These signals were normalized before being provided to the neural network. The objective was to reduce differences caused by absolute position or scale and allow the model to focus primarily on the writing dynamics and shape of each signature.

The preprocessing pipeline can therefore be summarized as:

Raw MCYT signature → Signal extraction → Resampling → Normalization → Neural network input

A target sequence length of 256 points was used during the autoencoder experiments.

This preprocessing stage was particularly important because the quality of the generated signatures directly depended on the representation learned from the original samples.

5.3 AUTOENCODER ARCHITECTURE

An autoencoder was selected to learn a compact representation of the online signatures.

The model consists of two main components: an encoder and a decoder. The encoder transforms the input signature into a lower-dimensional latent representation, while the decoder attempts to reconstruct the original sequence from this representation.

For an input sequence (X), the process can be represented as:

$$z = E(X)$$

where (E) represents the encoder and (z) is the latent representation.

The decoder then reconstructs the signature:

$$\hat{X} = D(z)$$

where (D) represents the decoder and (hat{X}) is the reconstructed signature.

The model was trained by minimizing the Mean Squared Error (MSE) between the original and reconstructed sequences:

$$\mathcal{L}_{MSE} = \frac{1}{N} \sum_{i=1}^{N} \left(X_i - \hat{X}_i \right)^2$$

The sequential autoencoder used a hidden dimension of **128** and a latent dimension of **32**. The Adam optimizer was used with a learning rate of 10^(-3).

Several numbers of training epochs were investigated during the development phase. Initial experiments were performed with different configurations before selecting values of approximately 100 to 150 epochs, depending on the experiment.

The evolution of the reconstruction loss was monitored during training to ensure that the model successfully learned to reproduce the main characteristics of the original signatures.

5.4 SYNTHETIC SIGNATURE GENERATION

Once the autoencoder had been trained, synthetic signatures were generated by modifying the representation learned by the model.

The basic principle consisted of encoding a genuine signature into the latent space, adding a controlled random perturbation, and decoding the modified latent vector:

$$z' = z + \varepsilon$$

where (z) is the original latent representation and (epsilon) represents Gaussian noise.

The synthetic signature is then obtained as:

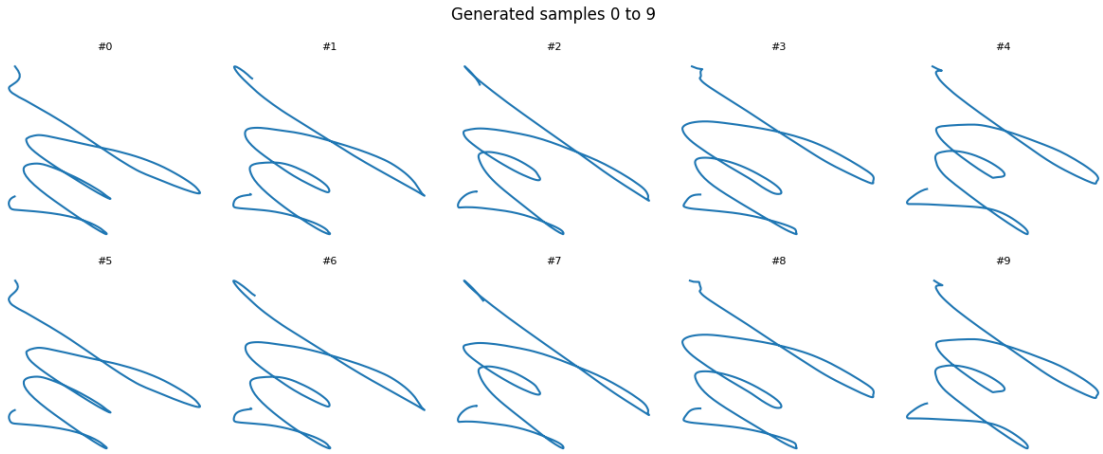
$$X_{\text{synthetic}} = D(z')$$

The amount of noise introduced during generation plays an important role. If the noise is too small, the generated signature remains extremely close to the original sample and provides little additional variability. Conversely, excessive noise can modify the signature too strongly and remove the biometric characteristics of the original user.

Several noise levels were therefore tested during the internship. Values ranging from approximately 0.02 to 3.0 were explored.

The first experiments showed that low noise values generated signatures with very small DTW distances from the genuine samples. Although these signatures were successfully recognized as belonging to the correct user, they were often too similar to the original sample to provide meaningful data augmentation.

A noise standard deviation of 3.0 was finally retained for the main experiments. This value provided a better compromise between similarity to the original signature and variability among generated samples.



5.5 Initial Evaluation Using Dynamic Time Warping

The first evaluation stage focused on determining whether the generated signatures remained biometrically similar to genuine samples.

Dynamic Time Warping was used to calculate the distance between each tested signature and the reference model of the corresponding user **[6].**

For each user, the reference DTW model was initially constructed using five genuine signatures: v01,v02,v03,v04 and v05.

The genuine test signatures (v06) to (v10) were then compared with this model.

Three categories of impostor or generated samples were also evaluated:

1. Random forgeries: genuine signatures belonging to other users;
2. Skilled forgeries: imitation signatures available in MCYT;
3. Synthetic signatures: signatures generated by the autoencoder.

For a test signature, the DTW distance to the user's reference model was calculated. A low distance indicates strong similarity to the enrolled user, whereas a high distance indicates a different writing pattern.

This experiment made it possible to compare the distribution of distances obtained for genuine signatures, random forgeries, skilled forgeries and synthetic signatures.

The DTW implementation was first validated by reproducing a reference baseline on the complete MCYT dataset. The obtained identification performance and verification metrics confirmed that the evaluation pipeline behaved correctly before synthetic samples were introduced.

5.6 EVALUATION OF SYNTHETIC SIGNATURES WITH DTW

After validating the DTW baseline, the synthetic signatures were evaluated as potential impostor samples.

The purpose of this experiment was to determine whether the generated samples were sufficiently similar to genuine signatures to be accepted by the biometric system.

For each user, an autoencoder was trained and five synthetic signatures were generated. The following four distributions were then compared:

- genuine signatures;
- random forgeries;
- skilled forgeries;
- synthetic signatures.

The analysis was performed using the DTW distances as well as biometric verification metrics.

A particularly important experiment consisted of applying the decision threshold determined using the skilled forgeries to the synthetic signatures. This made it possible to calculate the proportion of generated signatures that would be accepted as genuine by the DTW verification system.

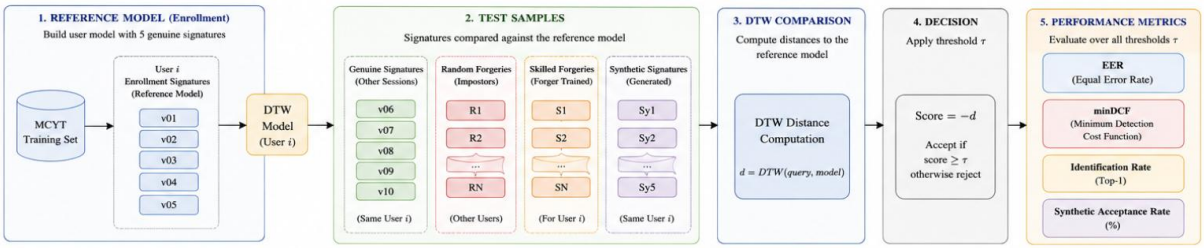


Figure 1: DTW evaluation protocol

This stage provided an initial quantitative evaluation of the biometric realism of the generated samples and made it possible to determine whether synthetic signatures could be accepted as genuine by the DTW-based verification system.

Based on these results, the next stage investigated a different question: whether synthetic signatures could be used as enrollment samples to compensate for a limited number of genuine signatures.

For this reason, a second experimental stage was introduced.

5.7 ENROLLMENT EXPERIMENTS WITH DTW

To investigate whether synthetic signatures could reduce the number of genuine enrollment samples, several configurations were compared.

The first configuration used only: **1 genuine signature** (corresponding to: train= v01 and test= v06-v10 + skilled + random)

The second configuration corresponded to the reference scenario: **5 genuine signatures** (corresponding to: train= v01-v05 and test= v06-v10 + skilled + random)

A third configuration used: **1 genuine signature + 4 synthetic signatures** (corresponding to: train= v01 + 4synthetic of v01 and test= v06-v10 + skilled + random)

Finally, an additional experiment increased the number of generated samples to: **1 genuine signature + 10 synthetic** signatures (corresponding to: train= v01 + 10synthetic of v01 and test= v06-v10 + skilled + random)

In the synthetic configurations, the generated signatures were derived from the single genuine enrollment sample. Importantly, synthetic signatures were used only during enrollment and not as genuine test samples.

The genuine test set remained composed of signatures (v06) to (v10), while random and skilled forgeries were used to evaluate the security of the system.

These experiments provided a first answer regarding the usefulness of synthetic enrollment data. However, the limited improvement observed with DTW suggested that a learning-based classifier could be more appropriate for evaluating data augmentation.

5.8 STATISTICAL FEATURE EXTRACTION

The next stage of the project replaced direct temporal comparison with a fixed-dimensional representation of each signature.

Each signature was transformed into a vector containing 230 statistical and frequency-domain features **[9].**

The feature representation included information derived from several signal families: pressure, horizontal and vertical positions, velocity, acceleration, derivatives of pressure, frequency-domain information obtained using the Discrete Cosine Transform (DCT) and global information related to sequence length.

For each signal, several statistical descriptors were calculated, including: mean, standard deviation, median, interquartile range, minimum, maximum, skewness and kurtosis.

DCT coefficients were additionally extracted to represent the global evolution of selected temporal signals in a compact form.

The resulting feature vector provided a fixed-size representation that could be directly used as input to conventional machine learning classifiers.

Before classification, the features were standardized in order to reduce the influence of differences in scale between descriptors.

5.9 RANDOM FOREST CLASSIFICATION

Random Forest was investigated as the first alternative to DTW **[7].**

A Random Forest combines multiple decision trees trained on different subsets of the available data and features. The final prediction is obtained by aggregating the predictions of the individual trees.

In this project, each signature was represented by the previously extracted feature vector, and the classifier was trained to distinguish between the different users of the MCYT database.

The purpose of this experiment was mainly exploratory: it allowed the complete feature-based classification pipeline to be validated before investigating a classifier specifically used in previous online signature verification research.

The same enrollment principle was maintained so that the effect of synthetic signatures could be compared with the previous experiments.

5.10 DIAGONAL LINEAR DISCRIMINANT ANALYSIS

The final classifier investigated during the internship was diagonal Linear Discriminant Analysis (diagLDA).

Linear Discriminant Analysis models each class according to its statistical distribution and searches for discriminative information capable of separating the different users. In the diagonal version, the covariance matrix is approximated using only its diagonal elements.

This simplification considerably reduces the number of parameters that need to be estimated and provides a computationally efficient classifier for high-dimensional feature representations.

For each user (c), a class mean vector (mu_c) is estimated from the available enrollment signatures. A shared diagonal within-class variance is also calculated from the training data.

For a feature vector (x), the discriminant score associated with class (c) can be expressed as:

$$g_c(x) = x^T \Sigma^{-1} \mu_c - \frac{1}{2} \mu_c^T \Sigma^{-1} \mu_c + \log P(c)$$

where (Sigma) represents the shared diagonal covariance matrix and (P(c)) is the prior probability of the class.

The user corresponding to the highest discriminant score is selected during identification.

This method was particularly relevant to the objective of the internship because, unlike DTW, its statistical model depends directly on the available training samples. Synthetic signatures could therefore potentially improve the estimation of the user's distribution when only one genuine sample was available.

5.11 Final Experimental Protocol

The final experiment compared three enrollment configurations using diagLDA.

Configuration 1 – One Genuine Signature:

Only the first genuine signature (v01) was used to represent each user.

This configuration simulated a very short enrollment procedure requiring only one signature from the user.

Configuration 2 – Five Genuine Signatures:

Five genuine signatures (v01) to (v05) were used for enrollment.

This configuration represented the conventional baseline and provided more information about natural intra-user variability.

Configuration 3 – One Genuine + Ten Synthetic Signatures:

The genuine signature (v01) was used together with ten synthetic signatures generated from this sample.

This configuration represented the main data augmentation experiment. Its purpose was to determine whether synthetic data could compensate for the absence of additional genuine enrollment signatures.

For all three configurations, the evaluation set remained unchanged. Genuine signatures (v06) to (v10) were used as positive samples. Random forgeries were obtained by considering genuine signatures from other users, while the skilled forgery samples were used to evaluate the system under a more difficult attack scenario.

Keeping the test protocol unchanged was essential to ensure a fair comparison between the different enrollment configurations.

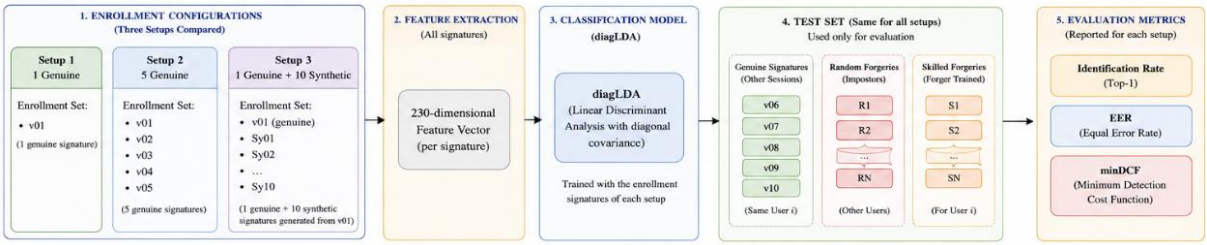


Figure 2: Final enrollment protocol

5.12 EVALUATION METRICS

Three main performance indicators were used throughout the experiments.

Identification Rate:

The Identification Rate measures the percentage of genuine test signatures assigned to the correct user:

$$IR = \frac{N_{\text{correct}}}{N_{\text{total}}} \times 100$$

A higher value therefore indicates better identification performance.

Equal Error Rate:

For verification experiments, two types of errors are particularly important.

The False Acceptance Rate (FAR) represents the proportion of impostor signatures incorrectly accepted as genuine.

The False Rejection Rate (FRR) represents the proportion of genuine signatures incorrectly rejected.

The Equal Error Rate corresponds to the operating point where:

$$FAR(\tau_{\text{EER}}) = FRR(\tau_{\text{EER}}) = EER$$

A lower EER indicates better separation between genuine users and impostors.

Minimum Detection Cost Function:

The minimum Detection Cost Function combines false acceptance and false rejection errors into a single measure. In the experiments performed during this internship, both error types were considered when determining the optimal operating threshold.

As with EER, a lower minDCF indicates better verification performance.

The metrics were calculated separately for random forgeries and skilled forgeries, since skilled attacks represent a significantly more challenging verification scenario.

5.13 SUMMARY OF THE EXPERIMENTAL PIPELINE

The complete methodology developed during the internship can be summarized in four main stages.

First, genuine MCYT signatures were preprocessed and used to train the autoencoder. The latent representation learned by the network was then perturbed to generate synthetic signatures.

Second, the generated signatures were evaluated using DTW to determine their similarity to genuine signatures and their ability to be accepted by a conventional verification system.

Third, the study investigated whether synthetic signatures could be used as enrollment samples by comparing different genuine/synthetic configurations.

Finally, the temporal signatures were converted into 230-dimensional feature vectors and evaluated using Random Forest and diagLDA classifiers. This final stage made it possible to assess whether synthetic data could compensate for a reduced number of genuine enrollment samples in a learning-based biometric system.

The following chapter presents and discusses the results obtained during these different experimental stages.

6. EXPERIMENTAL RESULTS AND DISCUSSION

This chapter presents the experimental results obtained throughout the internship. The experiments were conducted progressively in order to first validate the signature verification pipeline, then evaluate the biometric characteristics of the generated signatures, and finally determine whether synthetic samples could improve the enrollment process.

The analysis is divided into three main stages. First, Dynamic Time Warping (DTW) was evaluated on the MCYT database and used to study the similarity between genuine, forged and synthetic signatures. Second, synthetic signatures were introduced into the enrollment set and their effect on DTW performance was analyzed. Finally, feature-based statistical classifiers were investigated, with particular attention paid to diagLDA.

6.1 VALIDATION OF THE DTW BASELINE

Before evaluating synthetic signatures, it was necessary to verify that the implemented DTW pipeline provided reliable results on the MCYT database.

Five genuine signatures per user, from v01 to v05, were used as reference signatures, while v06 to v10 were reserved for genuine testing. Random and skilled forgeries were evaluated separately.

The baseline results are presented in Table 1.

Table 1 – DTW baseline results using five genuine enrollment signatures

Metric	Random Forgeries	Skilled Forgeries
EER (%)	0.7971	3.8788
minDCF (%)	0.5838	3.8121
Pmiss at minDCF	0.009697	0.041212
Pfa at minDCF	0.001978	0.035030

The identification rate obtained with this configuration was: Identification_Rate = 99.03%.

These results indicate that the DTW implementation provides very good discrimination between genuine signatures and forgeries.

Random forgeries are particularly easy to distinguish, with an EER below 1%. This result is expected because random forgeries correspond to genuine signatures produced by completely different users and therefore generally present very different trajectories and writing dynamics.

Skilled forgeries represent a more challenging situation. Their EER reaches approximately 3.88%, which remains low but is significantly higher than the random forgery result. This difference confirms that skilled forgeries reproduce part of the characteristics of the targeted signature and are therefore more difficult to detect.

Overall, these results provided a reliable baseline for the subsequent experiments involving synthetic signatures.

6.2 PRELIMINARY SYNTHETIC SIGNATURE EXPERIMENTS

Before generating synthetic signatures for the complete MCYT database, preliminary experiments were performed on a single user in order to study the behavior of the autoencoder and determine appropriate generation parameters.

Different values of latent noise were tested. With low noise values, the generated signatures remained extremely close to the original genuine sample. Their DTW distances were sometimes even lower than those observed between different genuine signatures.

Although this demonstrated that the autoencoder had successfully learned the structure of the input signature, such samples provided limited diversity and were therefore not ideal for data augmentation.

The latent noise was progressively increased from low values such as 0.02 up to 3.0. A noise standard deviation of 3.0 was eventually selected because it introduced more visible and dynamic variability while preserving the main characteristics of the user's signature.

For the selected user, the following DTW distances were obtained.

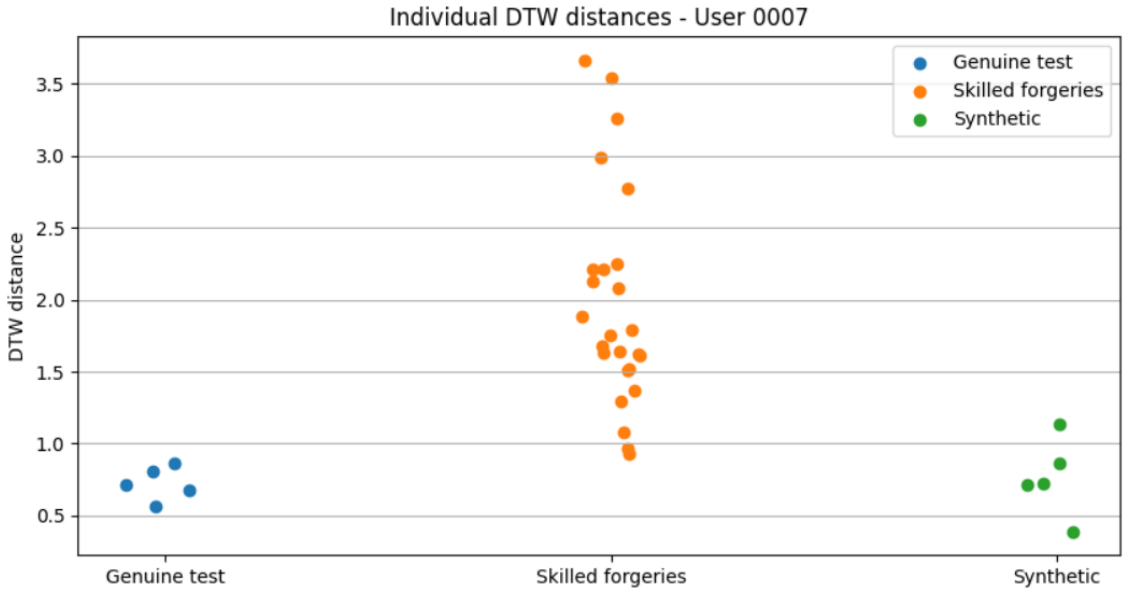


Figure 3: Individual DTW distances for genuine test signatures, skilled forgeries and synthetic signatures forthe selected user

Figure 3 shows the DTW distances obtained for three categories of signatures: genuine test signatures, skilled forgeries, and five synthetic signatures generated by the autoencoder.

A clear separation can be observed between genuine signatures and skilled forgeries. The genuine test signatures produce consistently low DTW distances, approximately between 0.56 and 0.86, with a mean distance of 0.723. In contrast, skilled forgeries show considerably higher

and more dispersed distances, ranging from approximately 0.93 to 3.66, with a mean value of 1.9755.

The synthetic signatures present a particularly interesting behavior. Their distances range from 0.3881 to 1.1386, with a mean value of 0.7652. This average is very close to the genuine mean distance of 0.723 and substantially lower than the skilled-forgery mean.

One synthetic signature even obtains a DTW distance of 0.3881, which is lower than the minimum distance observed among the genuine test signatures. Most of the remaining synthetic samples also fall within or close to the genuine distance range. Only one generated signature, with a distance of approximately 1.14, is clearly above the genuine test distribution, while still remaining close to the lower range of the skilled forgeries.

These preliminary results therefore suggested that the autoencoder was able to generate signatures that were biometrically close to genuine samples from the same user. At the same time, the dispersion observed among the synthetic signatures indicated that the latent noise introduced some variability rather than producing identical copies of the original signature.

However, this experiment involved only one user and a limited number of synthetic samples. It was therefore not sufficient to assess the general performance of the generation method. Based on these encouraging preliminary observations, the same evaluation was subsequently extended to the complete MCYT database.

6.3 EVALUATION OF SYNTHETIC SIGNATURES ON THE COMPLETE DATABASE

For the complete evaluation, the autoencoder was trained for each user and five synthetic signatures were generated per user using 100 training epochs and a noise standard deviation of 3.0.

The average DTW distances obtained for the four categories are presented in Table 2.

Table 2 – Global DTW distance summary

Signature type	Mean DTW distance
Genuine	0.8703
Synthetic	1.4694
Skilled forgery	3.3052
Random forgery	3.6927

A clear separation can be observed between genuine signatures and human forgeries.

Genuine signatures present the lowest average distance, with a value of 0.8703. Random forgeries produce the highest distance, at 3.6927, while skilled forgeries reach an average distance of 3.3052.

The synthetic signatures occupy an intermediate position, with an average distance of 1.4694.

Although their average distance is higher than that of genuine signatures, synthetic samples remain considerably closer to the genuine distribution than either type of human forgery.

This is an important observation because it indicates that the autoencoder has preserved a significant amount of user-specific information during generation.

6.3.1 Distribution Analysis

The distance distributions provide more information than the mean values alone.

The genuine signatures form a relatively compact distribution concentrated around low DTW distances. Random and skilled forgeries are shifted toward substantially higher distances.

Synthetic signatures partially overlap with the genuine distribution while presenting greater dispersion.

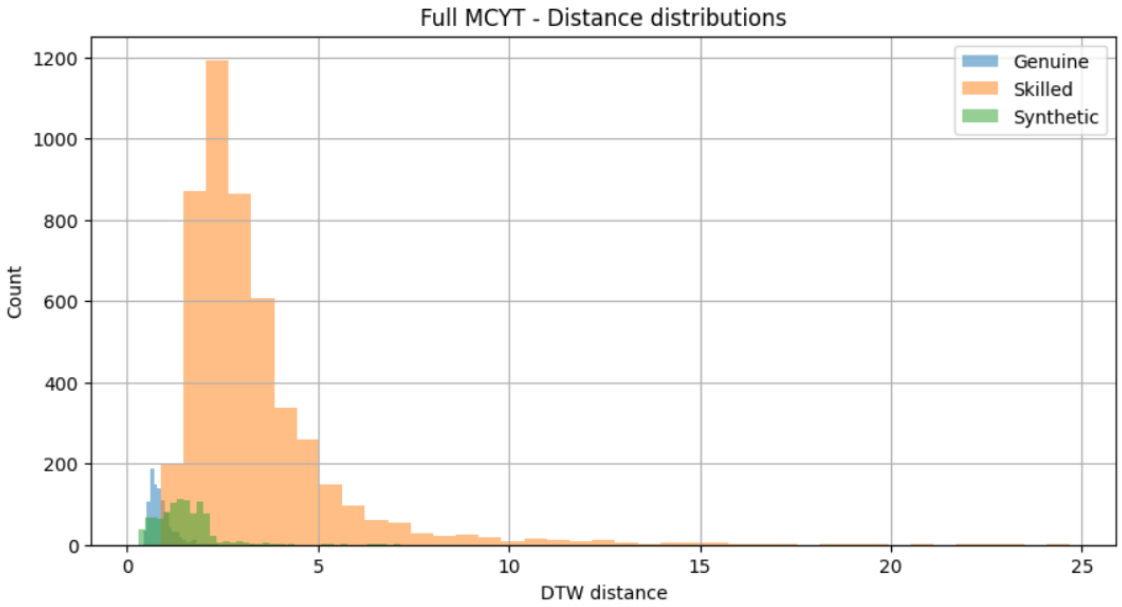


Figure 4: Distribution of DTW distances for genuine, random, skilled and synthetic signatures

This behavior suggests that the synthetic signatures preserve important biometric characteristics while introducing additional variability.

Nevertheless, the overlap between the genuine and synthetic distributions also raises an important security question: could synthetic signatures be accepted as genuine by the verification system?

6.4 Synthetic Signatures as Potential Forgeries

To investigate this question, the synthetic signatures were treated as potential attacks against the DTW verification system.

The resulting metrics are shown in Table 3.

Table 3 – Verification results for different forgery types

Type	EER (%)	minDCF (%)
Random forgeries	0.82	0.59
Skilled forgeries	4.31	4.08
Synthetic signatures	25.13	23.18

The results reveal a major difference between human forgeries and generated signatures.

The EER obtained for random forgeries is only 0.82%, while skilled forgeries reach 4.31%. In contrast, the EER obtained when synthetic signatures are considered as forgeries reaches 25.13%.

Similarly, the minDCF increases from 0.59% for random forgeries and 4.08% for skilled forgeries to 23.18% for synthetic signatures.

At first sight, the higher EER obtained for synthetic signatures could appear to indicate poorer synthetic quality. However, in this particular experiment, the opposite interpretation is more relevant. Because the synthetic signatures are considered impostor samples, a high EER means that the verification system has difficulty distinguishing them from genuine signatures.

The synthetic samples are therefore substantially more similar to genuine signatures than conventional human forgeries from the perspective of the DTW system.

6.5 SYNTHETIC SIGNATURE ACCEPTANCE RATE

A complementary experiment was performed using the decision threshold obtained from the skilled-forgery minDCF analysis.

The selected threshold was: Threshold = -1.462325.

At this operating point, the skilled forgery False Acceptance Rate was approximately: FAR_skilled = 3.86%.

The same threshold was then applied to the generated signatures.

The resulting synthetic acceptance rate was: Synthetic_Acceptance_Rate = 52.10%.

This means that more than half of the generated signatures were accepted as genuine by the DTW verification system.

This result represents one of the most significant findings of the first experimental stage. It confirms that the generated samples do not simply resemble signatures visually; they also reproduce characteristics that are relevant to the biometric comparison performed by DTW.

At the same time, this result must be interpreted carefully. A high acceptance rate demonstrates similarity to genuine signatures, but it does not necessarily mean that the generated samples provide useful additional information during classifier training.

This distinction motivated the following enrollment experiments.

6.6 SYNTHETIC SIGNATURES FOR DTW ENROLLMENT

The next objective was to determine whether synthetic signatures could compensate for a reduced number of genuine enrollment samples.

Four configurations were compared:

1. one genuine signature;
2. five genuine signatures;
3. one genuine signature + four synthetic signatures;
4. one genuine signature + ten synthetic signatures.

The results are summarized in Table 4.

Table 4 – DTW enrollment experiments

Setup	ID Rate (%)	EER Random (%)	EER Skilled (%)	minDCF Random (%)	minDCF Skilled (%)
1 Genuine	97.7576	1.5905	5.8788	1.3729	5.6727
5 Genuine	99.0303	0.7971	3.8788	0.5838	3.8121
1 Genuine + 4 Synthetic	97.5758	1.4545	6.4364	1.3021	6.3152
1 Genuine + 10 Synthetic	97.6364	1.4545	6.4969	1.3102	6.3696

The five-genuine-signature configuration clearly provides the best overall performance.

Using only one genuine signature reduces the identification rate from 99.03% to 97.76% and increases both the random and skilled forgery error rates.

Adding four synthetic signatures produces a small improvement for random forgeries compared with the one-genuine configuration. The random EER decreases from 1.59% to 1.45%. However, the skilled forgery EER increases from 5.88% to 6.44%.

Increasing the number of synthetic signatures from four to ten does not significantly improve the results. The identification rate remains approximately 97.6%, and the skilled EER slightly increases to 6.50%.

These results therefore indicate that simply increasing the number of synthetic reference signatures does not reproduce the benefit obtained from collecting several independent genuine signatures.

6.7 INTERPRETATION OF THE DTW ENROLLMENT RESULTS

The results obtained with DTW can be explained partly by the nature of the algorithm.

DTW is primarily a sequence matching method. It does not train a statistical model capable of learning a more accurate distribution when additional samples are introduced. Instead, the reference signatures are directly compared with the test sequence.

Consequently, increasing the number of synthetic reference samples does not necessarily improve the system in the same way that increasing the number of training examples would improve a machine learning classifier.

Furthermore, all synthetic signatures used in the reduced-enrollment experiments were generated from the same genuine sample, v01. They therefore represented variations around a single observation rather than independent observations of the user's natural handwriting variability.

This led to an important evolution of the project: synthetic data augmentation was subsequently evaluated using statistical classifiers whose parameters are directly estimated from the available training samples.

6.8 FEATURE-BASED CLASSIFICATION

The next stage replaced direct DTW comparison with the 230-dimensional feature representation described in Chapter 5.

A Random Forest classifier was first investigated as a faster feature-based alternative. This experiment served mainly to validate the transformation from variable-length temporal signatures to fixed-dimensional vectors and to explore whether a conventional machine learning classifier could benefit from synthetic enrollment samples.

The Random Forest experiments showed that the feature extraction and classification pipeline was operational. However, the final analysis focused on diagLDA, since this classifier was specifically relevant to the methodology investigated during the internship and provided a more direct statistical framework for studying the influence of the number of enrollment samples.

The Random Forest implementation and detailed results are therefore provided in Appendix A, while the following sections focus on the final diagLDA experiments.

6.9 DIAGLDA EXPERIMENTAL RESULTS

Three enrollment scenarios were evaluated using diagLDA:

- one genuine signature;
- five genuine signatures;
- one genuine signature + ten synthetic signatures.

The same genuine test samples and forgery protocols were used for all configurations.

The results are presented in Table 5.

Table 5 – Final diagLDA results

Setup	ID Rate (%)	EER Random (%)	EER Skilled (%)	minDCF Random (%)	minDCF Skilled (%)
1 Genuine	77.03	5.16	19.70	5.03	19.17
5 Genuine	96.91	0.55	10.19	0.48	10.19
1 Genuine + 10 Synthetic	74.12	6.76	21.08	6.70	20.10

The differences between the configurations are considerably larger than those observed with DTW.

The five-genuine configuration again achieves the best performance, reaching an identification rate of 96.91%.

When only one genuine signature is available, the identification rate decreases substantially to 77.03%. The skilled-forgery EER also increases from 10.19% to 19.70%.

This result illustrates the importance of having multiple genuine enrollment samples for statistical classification. Several genuine signatures allow the classifier to obtain a better representation of the natural variability of each user's handwriting.

6.10 IMPACT OF SYNTHETIC DATA ON DIAGLDA

The main objective of the final experiment was to determine whether ten synthetic signatures could compensate for the absence of four additional genuine enrollment samples.

The comparison between one genuine signature and one genuine signature augmented with ten synthetic samples is particularly informative.

The identification rate changes from 77.03% to 74.12%.

The random forgery EER changes from 5.16% to 6.76% and the skilled forgery EER changes from 19.70% to 21.08%.

Consequently, the addition of ten synthetic samples did not improve any of the main performance indicators.

This result is particularly interesting because the previous DTW experiments demonstrated that the generated signatures were biometrically similar to genuine samples. More than half of them could even be accepted by the DTW verification system.

However, being similar to a genuine signature is not equivalent to providing useful diversity for classifier training.

This distinction represents one of the main conclusions of the internship.

The synthetic samples were generated from a single genuine signature. As a consequence, they primarily explored variations around this particular observation. In contrast, five independently acquired genuine signatures naturally contain differences in trajectory, speed, pressure and execution that better represent the intra-user distribution.

Therefore, adding multiple correlated synthetic samples may increase the number of training observations without increasing the amount of useful biometric information.

6.11 COMPARISON BETWEEN DTW AND DIAGLDA

The experiments also highlight the different behavior of DTW and diagLDA when the number of enrollment signatures is reduced.

With DTW, the identification rate remained relatively high even with only one genuine signature: 97.76%.

With diagLDA, the same reduced-enrollment scenario resulted in 77.03%.

This difference can be explained by the fundamentally different nature of the two methods.

DTW directly compares the temporal structure of signatures and can therefore operate effectively with a small number of reference samples.

diagLDA, in contrast, relies on statistical information estimated from the training set. With only one genuine signature per user, the estimation of intra-user variability becomes extremely limited.

When five genuine signatures are available, diagLDA improves considerably and reaches an identification rate of 96.91%, demonstrating that the method benefits strongly from additional independent training samples.

This observation also explains why synthetic augmentation was initially considered promising for statistical classifiers: in principle, additional generated samples could improve the estimation of each user's distribution. In the experiments performed here, however, the generated signatures did not contain enough independent variability to provide this benefit.

6.12 MAIN FINDINGS

The experiments conducted during this internship led to several important observations.

First, the DTW baseline achieved very strong performance on MCYT, confirming that the implemented evaluation protocol was reliable.

Second, the autoencoder successfully generated signatures that remained close to the genuine biometric distribution. The mean synthetic DTW distance was substantially lower than the distances obtained for random and skilled forgeries.

Third, 52.10% of the synthetic signatures were accepted as genuine when evaluated using the decision threshold obtained from skilled forgeries. This demonstrates that the generated samples preserve important user-specific characteristics.

However, the enrollment experiments revealed a different result. Neither DTW nor diagLDA benefited significantly from replacing genuine enrollment signatures with synthetic ones.

With diagLDA, synthetic augmentation even resulted in a moderate degradation of performance compared with using only one genuine signature.

These findings demonstrate that two objectives must be distinguished when evaluating synthetic biometric data:

1. generating samples that are sufficiently realistic to resemble genuine signatures;
2. generating samples that contain sufficiently diverse and complementary information to improve classifier training.

The proposed autoencoder performed relatively well regarding the first objective but did not fully satisfy the second.

6.13 LIMITATIONS AND FUTURE IMPROVEMENTS

Several limitations of the present study should be considered.

The main limitation concerns the generation strategy. In the reduced-enrollment experiments, all synthetic samples were generated from only one genuine signature. The autoencoder therefore had very limited information about the natural variability of the user.

A more advanced generative model trained across multiple users could potentially learn general patterns of handwriting variability and transfer this knowledge when generating samples for a user with limited enrollment data.

Alternative generative architectures could also be investigated, including variational autoencoders, generative adversarial networks, or more recent generative approaches designed for sequential data.

Another possible improvement would consist of controlling the generation process using explicit biometric characteristics. Instead of introducing only random latent noise, variations in velocity, pressure, duration or trajectory could be generated independently. This could provide more realistic and interpretable intra-user variability.

The number and quality of synthetic samples could also be selected automatically. Rather than adding every generated signature to the enrollment set, a filtering mechanism could retain only samples that are sufficiently different from the original signature while remaining compatible with the genuine user distribution.

Finally, the experiments could be extended to additional online signature databases to determine whether the observations obtained on MCYT generalize to different acquisition devices, populations and signing conditions.

6.14 CHAPTER CONCLUSION

This chapter evaluated the synthetic signature generation approach under several biometric verification scenarios.

The DTW experiments demonstrated that the generated samples preserve substantial similarity to genuine signatures. Their average distance remained much lower than that of

human forgeries, and more than half of the synthetic samples were accepted as genuine under the selected DTW decision threshold.

However, the enrollment experiments showed that similarity alone is not sufficient for effective data augmentation. Five independently acquired genuine signatures consistently provided better performance than enrollment configurations based on a single genuine signature augmented with synthetic samples.

The final diagLDA experiment reinforced this conclusion. While five genuine signatures achieved an identification rate of 96.91%, adding ten synthetic signatures to a single genuine sample resulted in an identification rate of 74.12%, compared with 77.03% using the genuine sample alone.

The main limitation therefore appears to be the lack of sufficiently diverse intra-user information in the generated samples. These results provide a clear direction for future work: synthetic signature generation should not only aim to reproduce the identity of a writer, but also to model the natural variability that exists between multiple genuine executions of the same signature.

7. SUSTAINABLE DEVELOPMENT AND SOCIAL RESPONSIBILITY (DDRS)

7.1 ENVIRONMENTAL IMPACT

This internship mainly involved computational experiments using machine learning and deep learning models. The main environmental impact of the project therefore comes from the electricity consumption required for model training, particularly when using GPU resources.

Some experiments, especially those performed on the complete MCYT database, required several hours of computation. To reduce unnecessary calculations, preliminary tests were first performed on a limited number of users before extending the experiments to the full dataset.

Intermediate results, synthetic signatures and extracted features were also saved in order to avoid recomputing them after interruptions or disconnections. This approach helped reduce both computation time and energy consumption.

7.2 SOCIETAL AND ETHICAL ASPECTS

The project deals with biometric data, which raises important privacy and security issues.

Online signatures contain behavioral information such as trajectory, pressure and writing dynamics. These characteristics can be used for identity verification and therefore require careful handling and protection.

Synthetic biometric data also raises ethical questions. While generated signatures can be useful for data augmentation, they may also be sufficiently realistic to challenge authentication systems. During the experiments, more than half of the synthetic signatures were accepted as genuine under a specific DTW decision threshold.

This result illustrates the importance of considering not only the performance of generative models, but also their possible security implications.

7.3 Contribution of the Project

One of the motivations of this work was to investigate whether synthetic signatures could reduce the number of genuine samples required during enrollment.

A successful data augmentation approach could make biometric systems more convenient for users by reducing the number of signatures they need to provide.

However, the results showed that the generated signatures did not yet reproduce enough intra-user variability to replace several genuine samples effectively.

The project therefore highlights both the potential and the current limitations of synthetic data in biometric applications.

7.4 Possible Improvements

Several improvements could contribute to making this type of research more responsible and sustainable.

From an environmental perspective, experiments could continue to be optimized by reusing intermediate results, reducing unnecessary training runs and favoring computationally efficient models when possible.

From an ethical perspective, biometric datasets should remain anonymized and securely stored. Future work on synthetic signatures should also consider both their use for data augmentation and their potential use for testing the robustness of authentication systems.

Overall, responsible research in biometric artificial intelligence should consider not only accuracy, but also computational efficiency, privacy, security and user convenience.

SKILLS ASSESSMENT

This internship, allowed me to develop both technical and professional skills. Working on a research project was different from the academic projects I had previously completed. Instead of following a predefined procedure with an expected result, I had to progressively understand a research problem, propose solutions, analyze the results and adapt the methodology when an approach did not produce the expected outcome. This experience therefore helped me become more autonomous and develop a more critical approach to technical work.

TECHNICAL AND SCIENTIFIC SKILLS

One of the main areas in which I progressed was data analysis and machine learning. At the beginning of the internship, I first had to understand the MCYT database and the structure of online handwritten signatures. Unlike conventional datasets containing fixed-dimensional observations, online signatures are temporal sequences containing information such as position and pressure. Working with this type of data required me to understand how raw signals can be preprocessed, normalized and transformed before being used by different algorithms.

I also developed my knowledge of Dynamic Time Warping (DTW)**.** I learned how this algorithm can compare temporal sequences with different durations and how it can be applied to biometric signature verification. More importantly, I progressively understood its advantages and limitations. DTW provided very good results for direct signature comparison, but the experiments also showed that it was not necessarily the most appropriate method for evaluating the benefits of synthetic data as additional training samples. This observation eventually led to the exploration of statistical classifiers.

The development of the autoencoder was another important part of my technical progression. I became more familiar with neural network architectures, latent representations, reconstruction loss and the influence of hyperparameters. Generating synthetic signatures also required experimentation with the amount of noise introduced into the latent representation. I learned that obtaining a visually or mathematically similar synthetic sample does not necessarily mean that this sample will be useful for training a biometric system.

Finally, I worked with feature-based classification methods, particularly Random Forest and diagLDA**.** This required transforming temporal signatures into fixed-dimensional feature vectors and understanding how statistical classifiers use training samples differently from DTW. Through these successive approaches, I developed a broader understanding of machine learning pipelines, from raw data preprocessing to feature extraction, model training and performance evaluation.

EXPERIMENTAL AND RESEARCH SKILLS

Beyond the technical aspects, this internship significantly developed my understanding of the scientific research process**.**

One of the most important lessons was that research does not follow a perfectly linear path. The methodology evolved several times according to the results obtained. The first objective was to generate synthetic signatures and evaluate their similarity using DTW. Once this was achieved, the question became whether these signatures could actually improve the enrollment process. The results then showed the limitations of DTW for this purpose, which led to the introduction of statistical classification methods.

I therefore learned to consider an unexpected or negative result as useful information rather than simply as a failure. For example, the final experiments showed that adding synthetic signatures did not improve the performance obtained with a single genuine enrollment signature. Although this was not the initially expected result, its analysis revealed an important limitation of the generation approach: the synthetic signatures were similar to genuine signatures but did not reproduce enough independent intra-user variability.

The project also taught me the importance of establishing a rigorous experimental protocol**.** Training and test samples had to remain clearly separated to avoid data leakage, and the same test sets had to be maintained when comparing enrollment configurations. I also became familiar with biometric evaluation metrics such as Identification Rate, EER, FAR, FRR and minDCF and learned how to interpret them rather than simply report their numerical values.

Another skill I developed was reproducibility. Some experiments required significant computation time, and Google Colab sessions could be interrupted. I therefore progressively improved the implementation by saving generated signatures and intermediate results so that

experiments could be resumed without repeating previous computations. This experience showed me the importance of organizing code and experimental results carefully, particularly in a research context.

CONCLUSION

The objective of this internship was to investigate the use of synthetic online handwritten signatures for biometric signature verification, with particular attention to their potential use as additional enrollment samples. The main motivation was to determine whether artificially generated signatures could reduce the number of genuine samples required from a user while maintaining satisfactory verification performance.

The work was carried out progressively. First, the MCYT database and the principles of online signature verification were studied. A verification pipeline based on Dynamic Time Warping (DTW) was implemented and validated, providing a strong baseline with an identification rate of 99.03% when five genuine signatures were used for enrollment.

An autoencoder was then developed to generate synthetic signatures. Different generation parameters were explored, particularly the amount of noise introduced into the latent representation. The DTW experiments showed that the generated samples preserved important characteristics of the original users. Their average DTW distance was 1.4694, compared with 0.8703 for genuine signatures, 3.3052 for skilled forgeries and 3.6927 for random forgeries. Furthermore, when the decision threshold obtained from skilled forgeries was applied, 52.10% of the synthetic signatures were accepted as genuine.

These results demonstrated that the generation model was able to produce signatures presenting a significant biometric similarity to genuine samples. However, the subsequent enrollment experiments revealed an important distinction between generating realistic samples and generating useful training samples.

With DTW, using five genuine enrollment signatures consistently provided better results than using one genuine signature augmented with four or ten synthetic signatures. This observation also highlighted a limitation of DTW for evaluating data augmentation, since DTW performs direct sequence comparison rather than learning a statistical model from the enrollment samples.

The project was therefore extended to feature-based statistical classification. A Random Forest classifier was first investigated before implementing diagonal Linear Discriminant Analysis (diagLDA) using a 230-dimensional feature representation extracted from each signature.

The final diagLDA experiments confirmed the importance of genuine intra-user variability. Using five genuine signatures resulted in an identification rate of 96.91%, compared with 77.03% when only one genuine signature was available. Adding ten synthetic signatures to this single genuine sample did not compensate for the missing genuine data, resulting in an identification rate of 74.12%. Similar degradation was observed for the EER and minDCF metrics.

The main conclusion of this work is therefore that the synthetic signatures generated with the proposed autoencoder are biometrically similar to genuine signatures but do not provide sufficient independent intra-user variability to replace multiple genuine enrollment samples. Since the synthetic samples were generated from a single genuine signature, they mainly represented variations around the same observation rather than the natural variability obtained from several independent signing attempts.

Nevertheless, these results provide useful information for future research. Improving the generation strategy could involve using more advanced generative architectures, learning handwriting variability across multiple users, or explicitly controlling characteristics such as trajectory, pressure, velocity and duration. A selection mechanism could also be introduced to retain only synthetic samples that are sufficiently different from the original signature while remaining representative of the same user.

Beyond the experimental results, this internship provided the opportunity to develop a complete research pipeline, from data preprocessing and deep learning-based generation to biometric evaluation and statistical classification. It also highlighted the iterative nature of experimental research: initial results obtained with DTW led to new hypotheses, different enrollment configurations and, finally, the exploration of alternative classification methods.

Overall, this work demonstrates both the potential and the current limitations of synthetic data augmentation for online signature verification. Generating signatures that resemble genuine samples is achievable, but producing synthetic data that effectively reproduces natural intra-user variability remains a more challenging problem and represents an important direction for future work.

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APPENDIX

Appendix A: Random Forest FFR at fixed FAR operating points

Setup	FRR Random @ FAR 1% (%)	FRR Skilled @ FAR 1% (%)	FRR Random @ FAR 0.1% (%)	FRR Skilled @ FAR 0.1% (%)
1 Genuine	43.15	82.00	67.76	96.79
5 Genuine	0.85	38.06	4.00	69.33
1 Genuine + 10 Synthetic	39.03	82.00	68.61	99.27

Appendix B : Mobilan Carbon data sheet

